MuTau Channel: Status Update

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1. Summary

This week I investigated a normalization issue identified by Jonathan in the MC samples with pileup proton mixing.

Key findings: - Found and fixed a bug in the proton merging code - Measured the proton acceptance from data: **P = 24.5%** - Clarified the complete weight structure for all samples

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2. The Proton Mixing Procedure

Why we need it

- MC samples (DY, ttbar) have no PPS proton information
- Real data has pileup protons from ~30-40 collisions per bunch crossing
- We add fake pileup protons to MC to match data

The proton pool

File: `proton_pool_2018.root`

- Contains real protons from 2018 PPS data
- Each proton has: arm (0 or 1) and xi (momentum loss)
- Created from data with signal triggers but no kinematic cuts
- Protons are essentially random pileup

Assignment algorithm

For each MC event:

1. **Always assign 1 proton per arm** (xi_arm1_1, xi_arm2_1)
2. **Probabilistically add 2nd protons:**

Config	Probability	Description
P11	8.0%	1+1 protons
P12	2.0%	1+2 protons
P21	2.0%	2+1 protons
P22	0.5%	2+2 protons

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3. The Bug Found

Original code (WRONG)

```
# weight was set to fixed value, discarding event_weight
weight = 0.13
```

The `event_weight` from phase1 (containing generator weights and muon scale factors) was **lost**.

Fixed code (CORRECT)

```
# Now multiplies event_weight by proton acceptance
weight = event_weight * 0.13
```

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4. Proton Acceptance Measurement

What is it?

The fraction of events with at least 1 proton detected on **both** arms.

My measurement

Sample: `Data_2018_UL_MuTau_nano_merged_proton_vars.root`

Category	Events	Fraction
No protons on either arm	1,849	26.5%
Protons only on arm 0	1,600	23.0%
Protons only on arm 1	1,813	26.0%
Protons on BOTH arms	**1,708**	**24.5%**

Total events: 6,970 **Measured acceptance:** P = 0.245 (24.5%)

Comparison with previous value

- Matteo's value: **0.13 (13%)**
- My measurement: **0.245 (24.5%)**

The difference may be due to different event selection or data sample.

Question: Which value should we use?

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5. Complete Weight Structure

Data

```
weight = 1.0 (no corrections)
```

Drell-Yan (DY)

```
Phase 1:     event_weight = generator_weight * muon_SFs
Proton mix:  weight = event_weight * 0.13
Plotting:    final = weight * 1.81
```

ttbar

```
Phase 1:     event_weight = 0.15 * muon_SFs  (0.15 already included!)
Proton mix:  weight = event_weight * 0.13
Plotting:    final = weight * 1.0
```

QCD

```
Data-driven from same-sign selection
weight = custom normalization
```

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6. Cross-Section Normalization

Sample	Factor	Formula	Where applied
DY	1.81	N_MC / (L * sigma)	Plotting
ttbar	0.15	N_MC / (L * sigma)	Phase 1

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7. Current Issues

Issue	Status
Proton mixing overwrites weights	FIXED
event_weight lost after mixing	FIXED
Proton acceptance value (0.13 vs 0.245)	TO DISCUSS
DY normalization in plotting	TO CHECK

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8. Next Steps

1. **Clarify proton acceptance value** with Jonathan/Matteo
2. **Re-run proton mixing** for DY and ttbar with fixed code
3. **Verify plotting code** applies correct normalizations
4. **Generate new comparison plots**
5. **Cross-check results** with Jonathan

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9. Questions for Discussion

1. Which proton acceptance should we use: 0.13 or 0.245?
2. What selection was used for Matteo's 0.13 measurement?
3. Should we re-measure from a different stage sample?

Backup: Code for Measuring Acceptance

```
// In ROOT
TFile* f = TFile::Open("Data_merged_proton_vars.root");
TTree* tree = (TTree*)f->Get("tree");

Long64_t total = tree->GetEntries();
Long64_t good = tree->GetEntries(
    "Sum$(proton_multi_arm==0)>=1 && Sum$(proton_multi_arm==1)>=1"
);

double P = (double)good / (double)total;
cout << "Acceptance P = " << P << endl;
```

Backup: Files Used

File	Description
fase0_data_mutau.py	Phase 0 processing
fase1_data.py	Phase 1 selection
fase1_dy.py	Phase 1 for DY
fase1_ttjets.py	Phase 1 for ttbar
merge_pp_mutau.py	Proton mixing

File	Description
plot_m.py	Plotting
check_proton_acceptance.py	Acceptance measurement